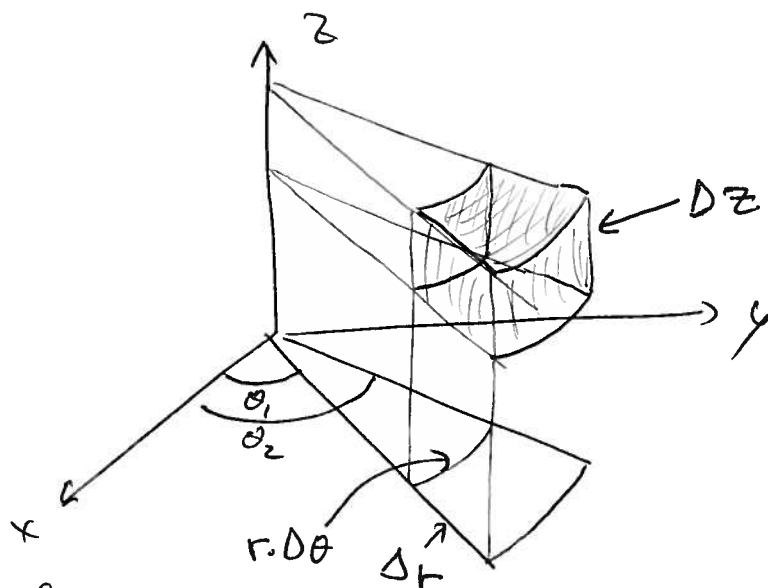


§ 13.8 Triple integrals in cylindrical and spherical coordinates

"Cylindrical box"

$$dA = r dr d\theta dz$$

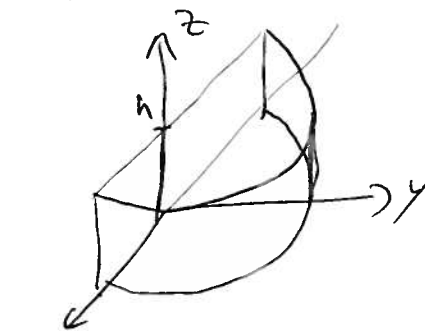


Ex Find volume & center of mass of half a cylinder of radius a & height h , & density ρ .

$$m = \int_0^h \int_0^\pi \int_0^a \rho r dr d\theta dz$$

$$= \int_0^h \int_0^\pi \left. \frac{\rho r^2}{2} \right|_0^a d\theta dz = \int_0^h \int_0^\pi \frac{\rho a^2}{2} d\theta dz$$

$$= \int_0^h \frac{\rho a^2 \pi}{2} dz = \rho \frac{a^2 \pi}{2} \cdot h$$



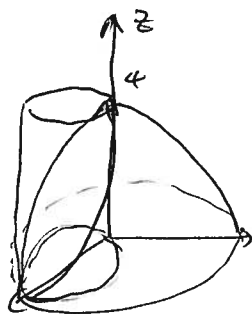
$$\bar{x} = 0 \quad \bar{z} = h/2$$

$$\bar{y} = \frac{1}{m} \int_0^h \int_0^\pi \int_0^a \rho r \cdot r \sin \theta dr d\theta dz = \frac{1}{m} \int_0^h \int_0^\pi \left. \frac{\rho r^3}{3} \sin \theta \right|_0^a d\theta dz$$

$$= \frac{1}{m} \int_0^h \int_0^\pi \frac{\rho a^3}{3} \sin \theta d\theta dz = \frac{1}{m} \int_0^h \left. -\frac{\rho a^3}{3} \cos \theta \right|_0^\pi dz = \frac{1}{m} \int_0^h \frac{2 \rho a^3}{3} dz$$

$$= \frac{1}{m} \cdot \frac{2 \rho a^3 h}{3} = \frac{4a}{3\pi}$$

Ex Volume under $z=4-x^2-y^2$, above $z=0$, inside $x^2+y^2 \leq 2x$



$$\begin{aligned} x^2 + y^2 &\leq 2x \\ (x-1)^2 + y^2 &= 1 \\ r^2 &= 2r \cos \theta \\ r &= 2 \cos \theta \end{aligned}$$

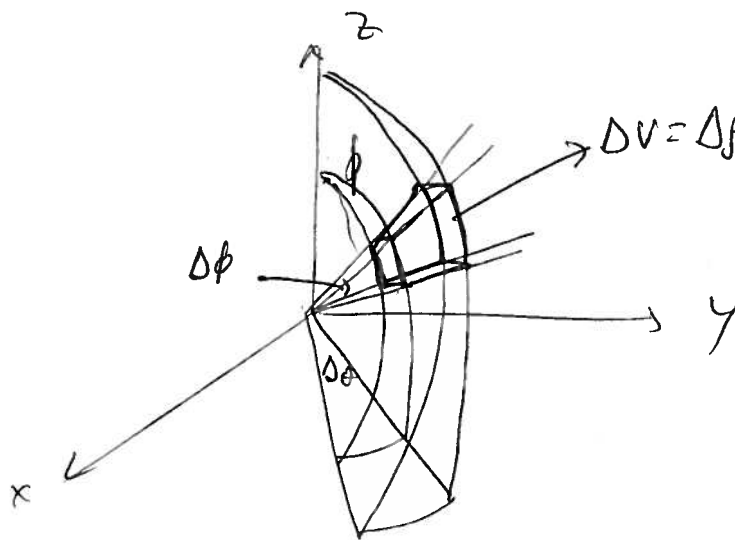
$$\iiint_S z \, dV = \int_{-\pi/2}^{\pi/2} \int_0^{2 \cos \theta} \int_0^{4-r^2} r \, dz \, dr \, d\theta$$

$$= \int_{-\pi/2}^{\pi/2} \int_0^{2 \cos \theta} r(4-r^2) \, dr \, d\theta = \int_{-\pi/2}^{\pi/2} \left[2r^2 - \frac{r^4}{4} \right]_0^{2 \cos \theta} d\theta$$

$$= \int_{-\pi/2}^{\pi/2} 8 \cos^2 \theta - 4 \cos^4 \theta \, d\theta = 8 \pi/2 - 4 \cdot \frac{3}{4} \frac{\pi}{2} = \frac{5\pi}{2}$$

Formula in book

"Spherical box"



$$\Delta V = \Delta \rho \cdot \rho \Delta \phi \sin \phi \Delta \theta$$

$$dV = \sin \phi \, \rho^2 \, d\rho \, d\theta \, d\phi$$

Ex Volume of a sphere of radius R

$$\begin{aligned} \iiint_S 1 \, dV &= \int_0^\pi \int_0^{2\pi} \int_0^R r^2 \sin\phi \, dr \, d\theta \, d\phi \\ &= \int_0^\pi \int_0^{2\pi} \left. \frac{r^3}{3} \right|_0^R d\theta \, d\phi = \int_0^\pi \int_0^{2\pi} \frac{R^3}{3} \sin\phi \, d\theta \, d\phi = \\ &= \int_0^\pi \frac{R^3}{3} \sin\phi \cdot 2\pi \, d\phi = \frac{2\pi R^3}{3} (-\cos\phi)_0^\pi = \frac{2\pi R^3}{3} (-\cos\pi + \cos 0) = \frac{4\pi R^3}{3} \end{aligned}$$

Ex Center of mass of hemisphere of radius R

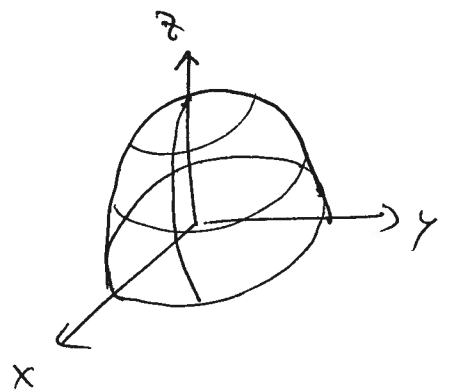
$$\bar{z} = \frac{1}{\frac{2\pi R^3}{3}} \iiint_{0 \leq \phi \leq \pi/2} r^2 \sin\phi \, r \cos\phi \, dr \, d\theta \, d\phi$$

$$= \frac{3}{2\pi R^3} \int_0^{\pi/2} \int_0^{2\pi} \left. \frac{r^4}{4} \right|_0^R \sin\phi \cos\phi \, d\theta \, d\phi$$

$$= \frac{3}{2\pi R^3} \int_0^{\pi/2} \int_0^{2\pi} \frac{R^4}{4} \sin\phi \cos\phi \, d\theta \, d\phi$$

$$= \frac{3}{2\pi R^3} \int_0^{\pi/2} \frac{2\pi R^4}{4} \sin\phi \cos\phi \, d\phi = \frac{3}{2\pi R^3} \frac{2\pi R^4}{4} \left. \frac{\sin^2\phi}{2} \right|_0^{\pi/2}$$

$$= \frac{3R}{4} \cdot \frac{1}{2} = \frac{3}{8} R$$



Ex Compute $\int_0^\pi \int_0^{2\pi} \int_0^a r^2 \sin\phi \, dr \, d\theta \, d\phi$ & $\int_0^\pi \int_0^{\sin\theta} \int_0^2 r \, dz \, dr \, d\theta$

Ex Find the volume of the solid bounded by $x^2 + y^2 + z^2 = 9$ below $z=0$ and inside $x^2 + y^2 = 4$ (also the solid bounded above by $z=xy$ below by $z=0$ & inside $x^2 + y^2 = 2x$)

Ex Find the center of mass of a solid hemisphere of radius R and density proportional to the distance from the center.